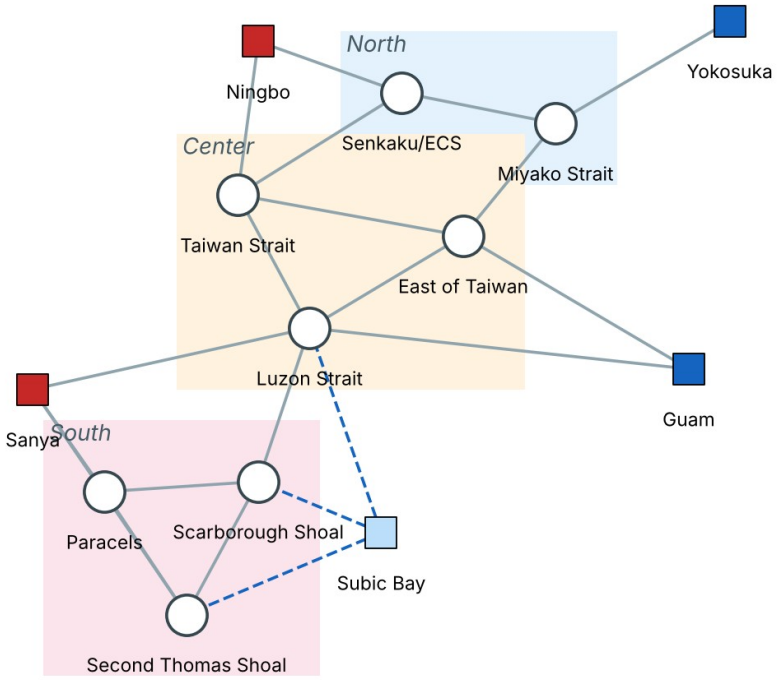
Gray-Zone Maritime Control in the First Island Chain

A networked escalation game | Workshop 1

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The problem: control without war



Network of 8 contested zones and 4 home ports

Players. Blue: 10 U.S. destroyers (DDG). Red: 8 navy ships (PLAN) + 6 coast guard / militia hulls (CCG).

Each period (2 days). Each side chooses where its ships go and whether to patrol (P) or assert (A: shadowing, close maneuvering, querying and warning; for the U.S. also freedom-of-navigation operations).

Control of a zone follows the local force ratio, so each side's best move depends on the other's.

Incidents can happen when both sides are present, and are likelier if someone asserts. Incidents raise an escalation level; at the top is war, which costs both sides.

Fog. Neither side sees the other's entire fleet disposition; only forces present where they also maintain forces. Storms can close zones.

Research questions and what changed since the proposal

RQ1

Control vs. war risk

Each side caps how much incident risk it will accept each period (its risk budget ε). How much more control does a side gain by accepting more risk, how much more likely does war become, and what risk level does each side pick when it knows the other side is picking too?

RQ2

Game theory vs. rules of thumb

Does planning with game theory (best responses, equilibrium or maximin plans) beat simple rules or random moves? Can both sides stay at patrol over many periods, because asserting once would end the truce and push both into riskier play from then on?

RQ3

Weather and redesign

Does the plan or Blue redesign that works best in normal weather still work best across calm, typhoon and extreme weather scenarios, both on average and in the worst scenario?

Feedback	What we did
Model too high level	Formal game on a graph
Q1 is a feasible region, use a MIP	Best response is an exact MIP; commander types become risk budgets ε
More ambitious questions	Budget-game equilibrium (RQ1), equilibrium play (RQ2), robustness and redesign (RQ3)
Define the resilient redesign	Three Blue designs
Propose values for missing data	Assumed values, listed in the report
Show preliminary results	Analytical results; 77 setups × 200 simulated games
Add weather (Prof. Sapsis)	Weather scenarios; expected and worst-case payoff

The game, one period at a time

1. The weather (sea state) of each region is drawn
2. Each side sees its own ships and detects some enemy ships
3. Ships low on endurance head home
4. Both sides choose moves and postures at the same time
5. Moves happen within movement budgets
6. Control updates; incidents are drawn; escalation level moves
7. Payoffs are paid; stop at war or after 60 periods

Control of zone v

$c \leftarrow (1 - \lambda)\cdot c + \lambda\cdot\varphi,\ \lambda = 0.3,\ \ \varphi = (S_B - S_R) \,/\, (S_B + S_R + \kappa)$

Presence builds control slowly; leaving lets it decay.

Incident chance

$h = (h_0 + h_A\cdot[\text{asserts}]) \cdot \min(S_B, S_R) \cdot (1 + \ell/2),\ \rho = 1 - e\text{^}(-h)$

Needs both sides present; grows with assertion and with the escalation level ℓ.

Payoff

$g = \Sigma\, V\cdot c - \gamma\cdot\text{moves} - \eta\cdot\ell - \zeta_D,\ \ \text{minus } W = 200 \text{ at war}$

Value of control minus fuel, tension and Blue's redesign cost ζ_D. War hurts both sides, so the game is not zero-sum.

Takeaway Every rule is explicit, so the same game can be played thousands of times by different strategies.

How the players decide

Best response as a MIP

Each period, choose moves and postures to maximize value of control minus costs, subject to the movement budget, endurance, and an escalation-risk budget:

expected incidents ≤ ε

Conservative / moderate / aggressive = ε of 0.05 / 0.15 / 0.40

Template game

Abstract each side to 14 templates: how to split ships North / Center / South, and patrol or assert.

A 14×14 strategic-form game

Solved with iterated elimination of strictly dominated strategies (IESDS), Nash equilibrium, maximin LP, correlated-equilibrium LP

Policies we compare

Random moves
Rule of thumb (rule-based)
Quick best response (greedy)
Best response (MIP)
Nash equilibrium plan (equilibrium)
Safety-first plan (maximin)

Same random draws and weather for every policy

Takeaway The proposal's strategy types are now one number, ε, inside an optimization (a MIP).

Small example: when does it pay to assert?

Blue \ Red	Red patrols	Red asserts
Blue patrols	0 , 0	−9.6 , +1.2
Blue asserts	+0.3 , −10.5	**−11.4 , −11.3**

Payoffs (Blue, Red) with force splits fixed, relative to both patrolling

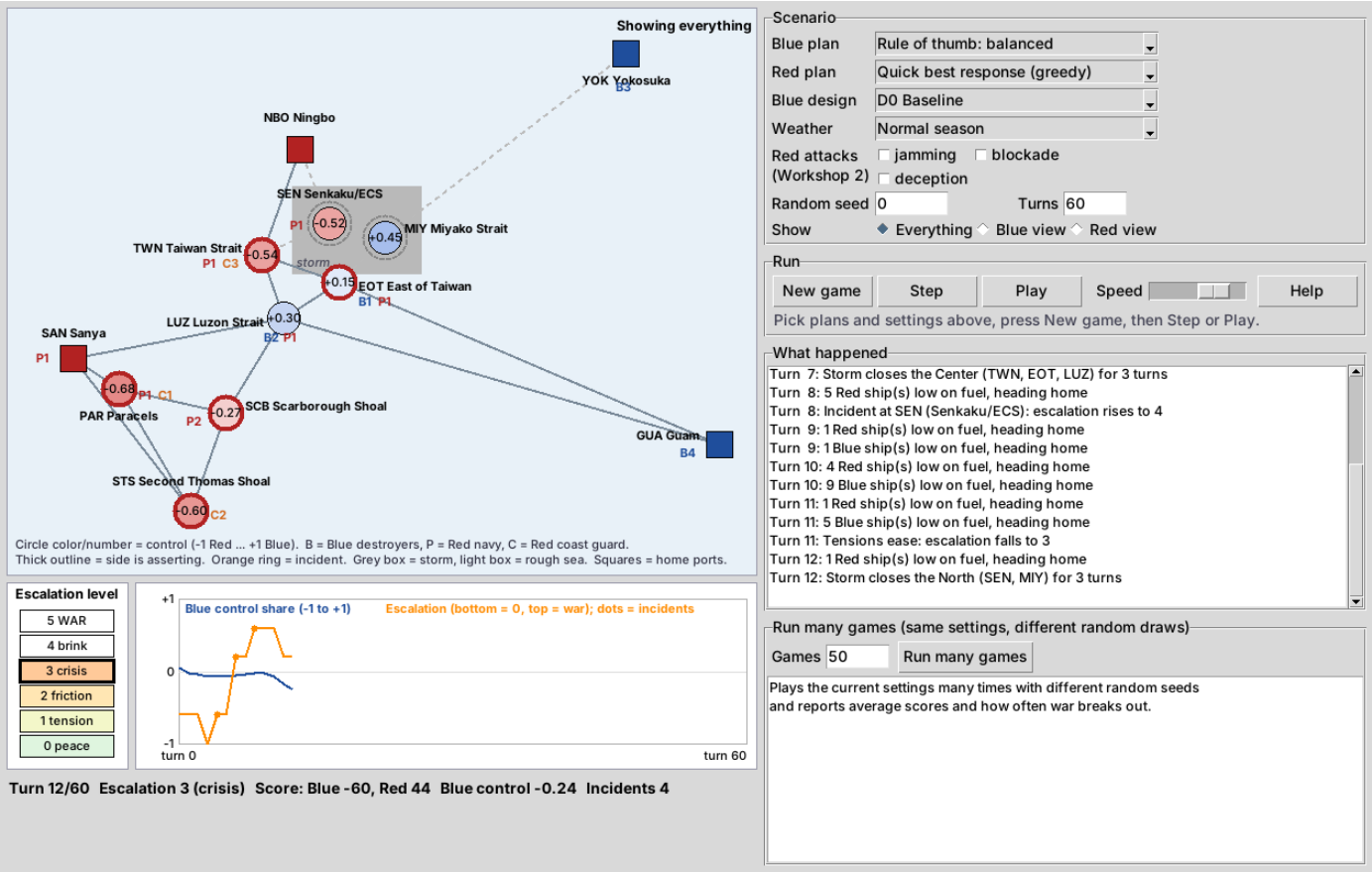
Best reply: assert if the other patrols, patrol if the other asserts. Two pure equilibria, plus a mixed one with mutual assertion 8% of the time.

A mediator helps: the best correlated equilibrium never recommends (assert, assert). A hotline can play that role.

Repetition helps: if both patrol and any assertion makes both fall back to the mixed equilibrium for good, mutual patrolling is a subgame perfect equilibrium for any discount factor above 0.18. Ours is 0.98.

Takeaway Restraint is not stable in a one-shot encounter, but it can be sustained by repeated play or a coordination device.

The prototype: watch a game play out



Pick a plan for each side, a design and the weather on the right; press New game, then Step or Play. Help explains the map.

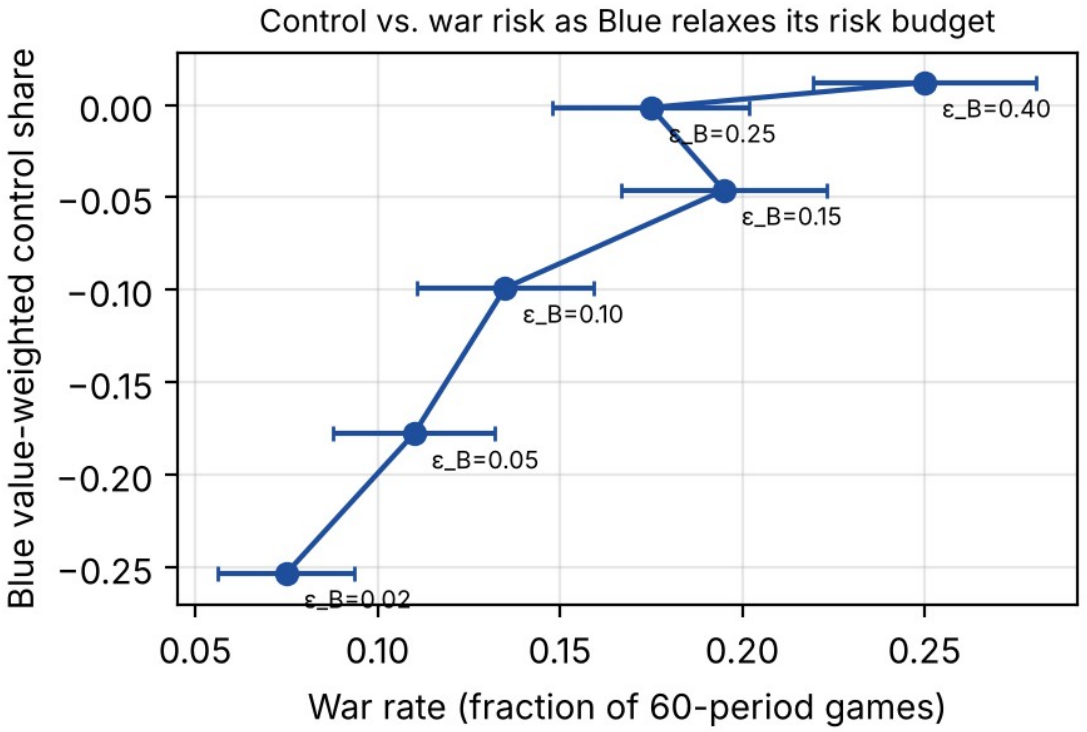
Zone color shows control: blue = Blue, red = Red.

B / P / C = destroyers, navy ships, coast guard. Thick outline = asserting.

Orange ring = incident. Shaded region = rough sea or storm.

Run many games plays the same setup as many simulated games and reports averages.

RQ1: more risk buys control, up to a point



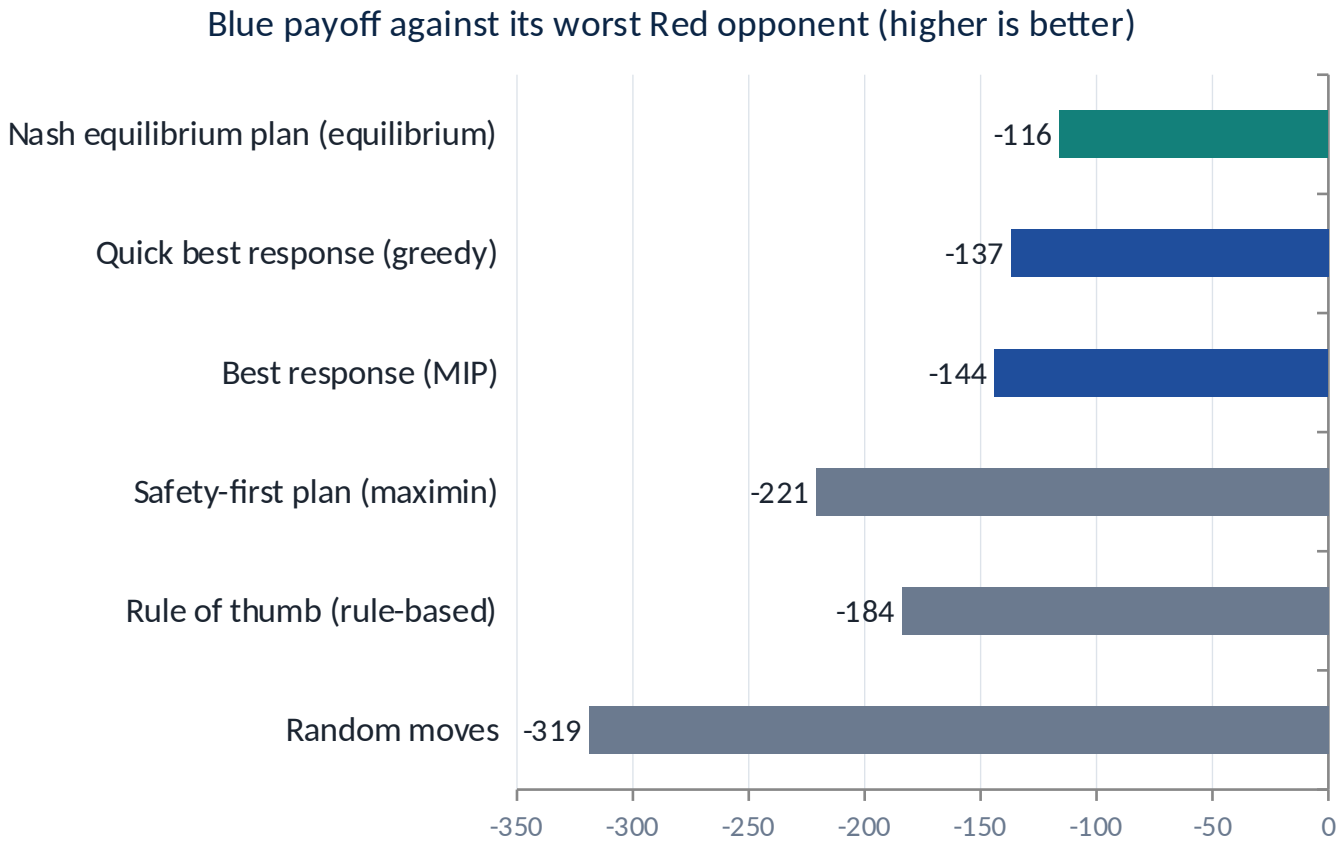
Raising Blue's budget from 0.02 to 0.15 lifts its control a lot; beyond 0.25 it barely moves.

Blue \ Red	Conservative	Moderate	Aggressive
Conservative	−194, 199 war 10%	−303, 315 war 11%	−308, 318 war 12%
Moderate	18, −36 war 15%	−144, 101 war 20%	−192, 140 war 21%
Aggressive	74, −80 war 14%	−83, 16 war 25%	**−151, 39 war 34%**

Risk-budget game: payoffs (Blue, Red) and war rate. Red bold = the only equilibrium.

Takeaway Both sides choose aggressive, giving the highest war rate (34%): a prisoner's dilemma.

RQ2: the Nash equilibrium plan is the safest bet against any Red



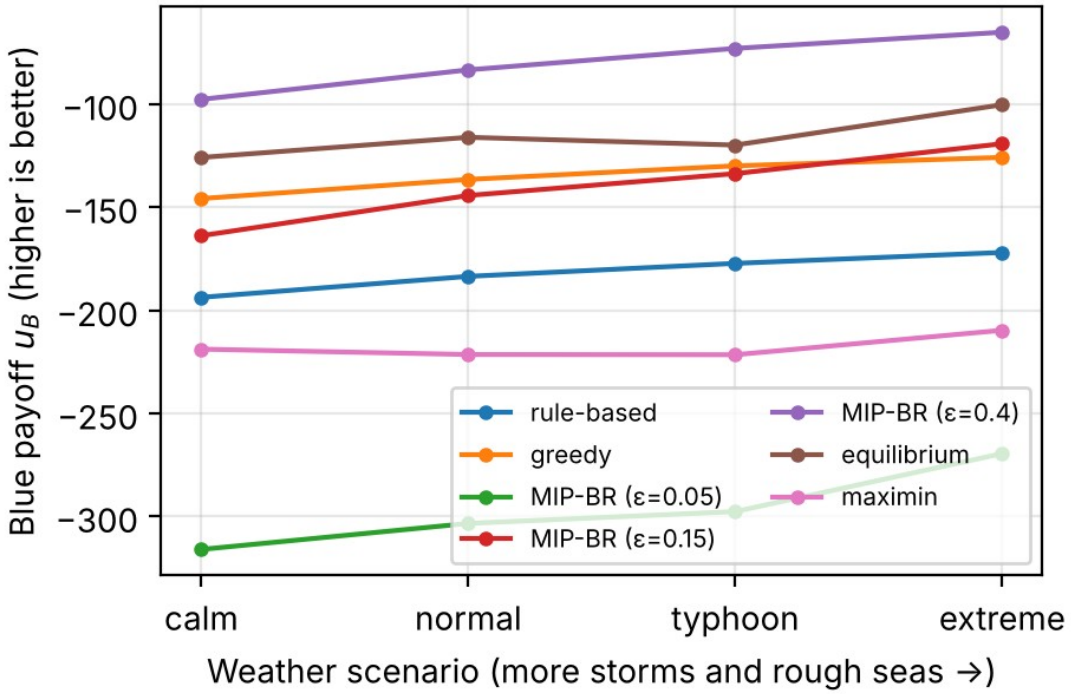
Nash equilibrium plan has the best worst case (−116) and half the war rate of the MIP best response against the same opponent (10% vs 20%).

Best responses (quick and MIP) win against weak opponents but escalate more.

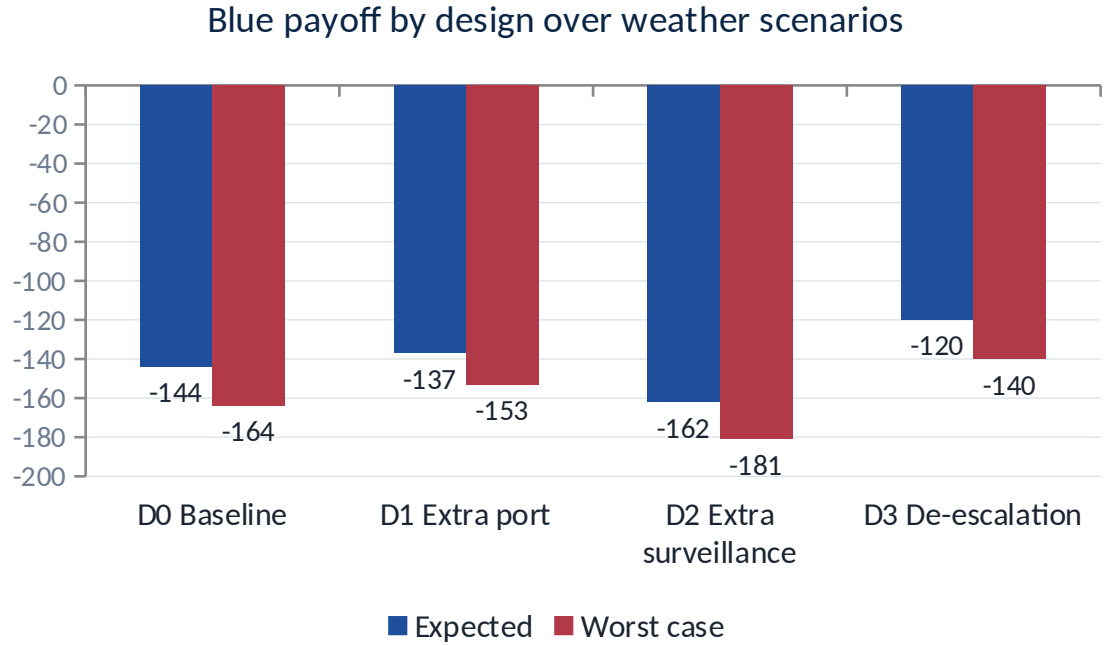
Safety-first plan is the most cautious (war ≤ 8%) but gives up the most control.

Quick vs MIP: the quick heuristic matches the MIP 67% of the time but breaks the risk budget 14% of the time.

RQ3: the best strategy holds up in bad weather



Rough seas weaken Red's coast guard hulls and storms pause operations, so most policies do slightly worse in calm seas. The top two plans stay on top in every scenario.



War rate (normal weather): D0 20%, D1 28%, D2 23%, D3 5%. De-escalation measures were given no cost yet.

Takeaway The plan that is best on average is also best in the worst weather. De-escalation measures are the best design and cut war about 4×.

What the evidence does and does not show

Supported

- Loosening the risk budget has sharply diminishing returns
- Left alone, both sides loosen their budgets (prisoner's dilemma)
- Restraint can be an equilibrium of the repeated game
- The Nash equilibrium plan is the most robust Blue policy here
- De-escalation measures are a joint gain under every weather scenario

Not yet shown

- All numbers are assumptions, not calibrated to data
- Policies look only one period ahead
- War penalty may be too small: the top-payoff policy also has the most wars
- De-escalation measures have no cost yet, so part of their win is built in
- Differences under about 20 payoff points are ties

Plan for Workshop 2



Beliefs and attacks

Bayesian beliefs about enemy positions; weather tied to the calendar (typhoon season, winter monsoon) with odds from climatology data; test the three coded Red attacks (blockade, jamming, false contacts); decide whether to keep the movement budget or drop it

Learning

Red agents that learn from experience: do they exploit predictable Blue play? An attacker that moves after seeing Blue

Redesign

Designs and combinations under attacks, persistent weather and a rare long typhoon; a cost for de-escalation measures

Report

Final report and presentation

Takeaway Next: learning, attacks and defenses, and resilient design, in that order.